IT, CE, AI -3 Extc-5

(3)

SVKM'S NMIMS

MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT ENGINEERING / SCHOOL OF TECHNOLOGY MANAGEMENT ENGINEERING

Program: B. Tech / MBA Tech (AI & DS/ AI & ML/ AI/CSE Cyber Security/Computer/IT/EXTC) Year: II/III Semester : III/V

AcademicYear: 2022-2023

Subject: Discrete Mathematics

Marks: 100

Time: 10.00 am - 1.00 pm

Date : 28 January 2023

Durations: 3 (Hrs)

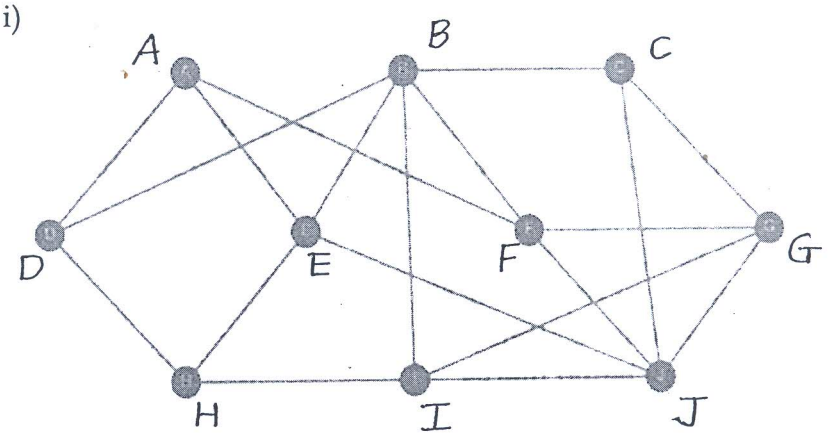
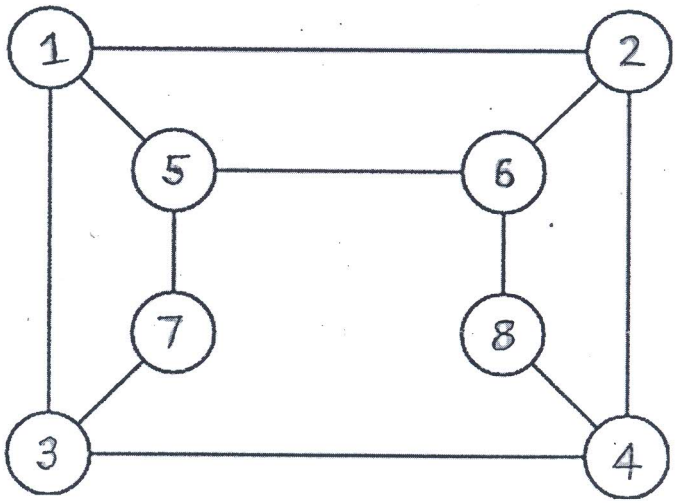
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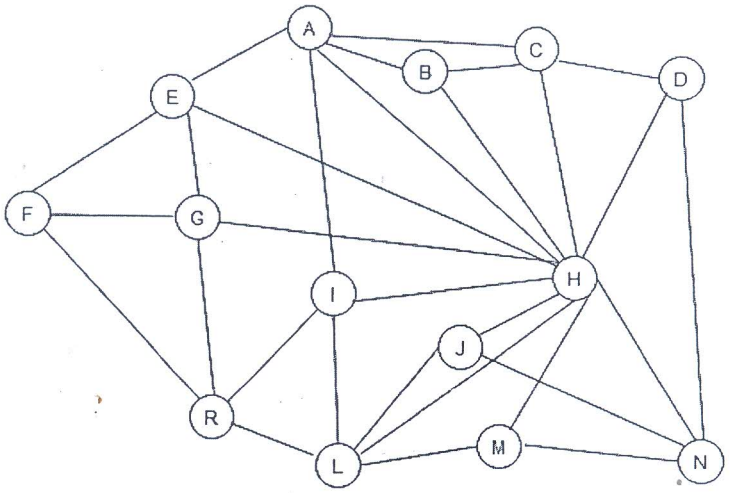
Re-Examination (2022-23)

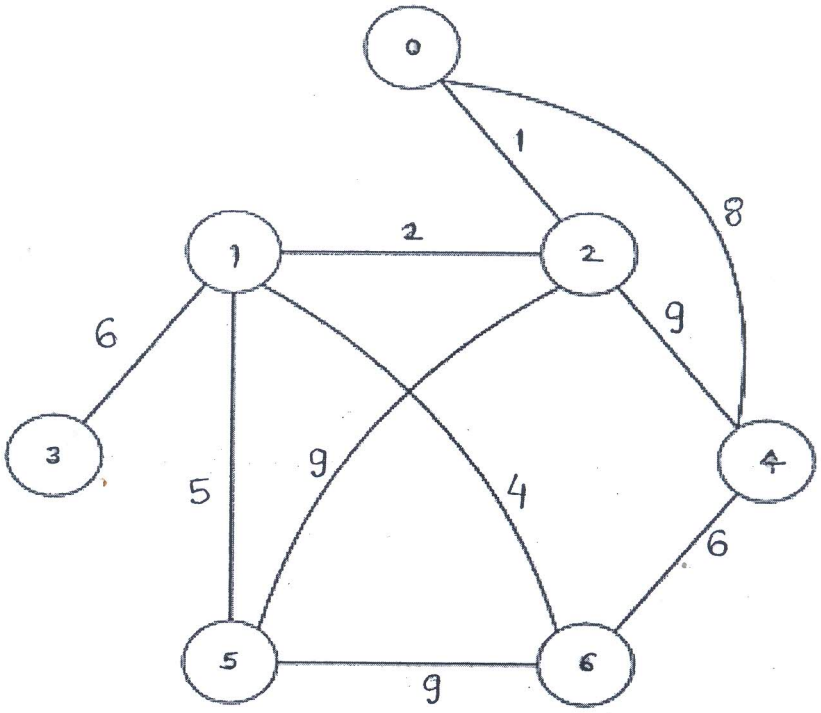
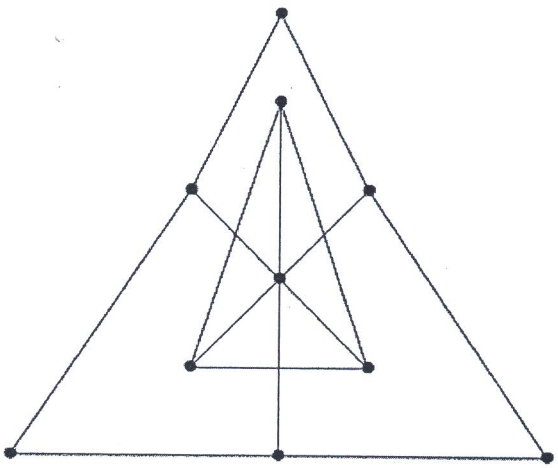
Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

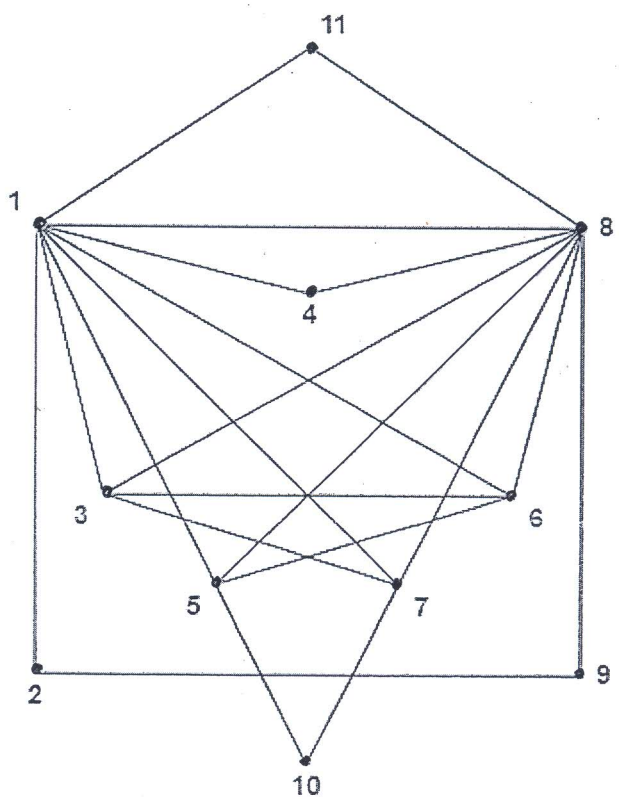
1. Questions 1 is compulsory.
2. Solve any four out of remaining six questions
3. **In all, 5 questions to be attempted**
4. All questions carry equal marks.
5. **Answer to each new question to be started on a fresh page.**
6. **Figures in brackets on the right hand side indicate full marks.**
7. **Assume Suitable data if necessary.**
8. **Use of scientific calculator is allowed.**

		Answer the following:	Marks
C01/SO2/BL3	Q.1 a.	From amongst 2000 literate individuals of a town, 70% read Marathi newspapers, 50% read English newspapers and 32.5% read both Marathi and English newspapers. Find the number of individuals, who read (a) at least one of the newspapers (b) only Marathi newspaper	[4]
C02/SO1/BL2	b.	Show that the statement $((p \vee q) \wedge (p \rightarrow r) \wedge (q \rightarrow r)) \rightarrow r$ is a tautology.	[4]
C01/SO1/BL1	c.	State pigeon hole principle. An elementary school has 400 students. Show that at least two of them are born on the same day of the year.	[4]

C04/SO1/BL3	d.	Check whether the following graphs exists. Justify your answer. i) A tree with 6 vertices and 6 edges. ii) A graph with degree sequence (0,1,1,2,3,3)	[4]
CO4/SO2/BL5	e.	Show that $\{\mathbb{Z}^+, *\}$ where $*$ is defined by $a*b=a$ for all $a,b\in\mathbb{Z}^+$ is a semi-group where $\mathbb{Z}^+$ is the set of all positive integers.	[4]
CO3/SO2/BL5	Q.2 a.	Determine the Homogeneous solution for the given recurrence relation $a_n-3a_{n-1}+2a_{n-2}=0$ with initial condition $a_0=5,\ a_1=3$	[6]
CO1/SO2/BL3	b.	Determine which of the following graphs are bi-partite. Justify your answer. i)  ii)  	[6]
CO3/SO2/BL4	c.	Let $\mathbb{Z}$ be a set of integers and let R be a relation on $A=\mathbb{Z}\times\mathbb{Z}$ defined by $(a,b)R(c,d)$ *if* $a+d=b+c$. Then prove that R is an equivalence relation.	[8]

CO2/SO1/BL2	Q.3 a.	Prove that $1 + 2 + 2^2 + 2^3 + \ldots\ldots\ldots + 2^n = 2^{n+1} - 1$ for all integers $n \geq 1$, using the principle mathematical induction.	[6]
CO2/SO2/BL6	b.	Estimate the chromatic number of the following graph using Welch-Powell Algorithm 	[6]
CO1/SO1/BL4	c.	Let f, g be functions from set of real number to set of real number such that $f(x) = 2x + 5$ and $g(x) = \frac{x}{3} - 7$. Find $f \circ g$ and $g \circ f$. Examine whether $f \circ g$ is bijective or not. If $f \circ g$ is bijective find inverse of $f \circ g$.	[8]
CO3/SO2/BL4	Q.4 a.	Prove that the set of real numbers $(\mathbb{R}, \oplus, \otimes)$ is a ring under the operations defined by $a \oplus b = a + b - 1$ & $a \otimes b = a + b - ab$	[6]
CO2/SO2/BL2	b.	Convert $q \vee (p \wedge r) \wedge \sim((p \vee r) \wedge q)$ into Disjunctive Normal Form.	[6]
CO3/SO1/BL5	c.	Evaluate the total solution of the recurrence relation $a_{n+2} - 5a_{n+1} + 6a_n = 6(5^n)$	[8]
CO2/SO1/BL5	Q.5 a.	G is a group of non-zero real numbers under multiplication and n is a positive integer. Check whether $f(x) = x^n$ is an isomorphism from G to G.	[6]
CO2/SO2/BL4	b.	Prove that $\sqrt{3}$ is an irrational number using method of contradiction.	[6]

CO3/SO1/BL3	c.	Construct a minimum spanning tree using Prims Algorithm. Also find minimum total weight. 	[8]
CO1/SO2/BL4	Q.6 a.	Let R be a relation on set of integers ($\mathbb{Z}$), such that aRb if and only if $a \leq b + 1$. Determine whether the relation R on the set $\mathbb{Z}$ is (1) irreflexive (2) antisymmetric (3) transitive with proper justification.	[6]
CO4/SO1/BL3	b.	Identify whether the following graph is planar or not. Give reason in detail. 	[6]

CO4/SO1/BL6	c.	Let G= {0, 1, 2, 3, 4, 5} be a set of congruence classes under addition modulo 6 i) Prepare composition table with respect to addition modulo 6 ii) Prove that G is an abelian group with respect to addition modulo 6 iii) Find the inverse of elements 2, 3 and 5. iv) Is it cyclic? v) Find the order of elements 2, 3 and sub groups generated by these elements.	[8]
CO2/SO2/BL3	Q.7 a.	Apply laws of logic to establish the equivalence: $\sim(p \wedge q) \rightarrow (\sim p \vee (\sim p \vee q)) \equiv (\sim p \vee q)$	[6]
CO3/SO7/BL2	b.	Solve the given recurrence relation to get the total solution: $a_n - a_{n-1} - 6a_{n-2} = 4$	[6]
CO4/SO1/BL3	c.	Determine whether Eulerian trail, Eulerian circuit, Hamiltonian path exists for the following graph. If so, find them.  	[8]